



Continuum Mechanics

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Chapter 01

Tensor Algebra

Part 5 of 5



Representation in a Coordinate Frame



Components of a Fourth-Order Tensor

Definition

Let $\mathbf{C} \in \mathcal{V}^4$ and suppose $\mathcal{F} = \{\mathbf{e}_i\}$ is a right-handed orthonormal frame. The 81 components of $\mathbf{C}$,

$$C_{ijkl} \in \mathbb{R}, \quad 1 \leq i, j, k, \ell \leq 3,$$

are defined by

$$C_{ijkl} = \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell)\mathbf{e}_j).$$



Proposition

Suppose that $\mathbf{C} \in \mathcal{V}^4$ and $U, T \in \mathcal{V}^2$ satisfy

$$U = \mathbf{C}(T).$$

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be a frame and write

$$U = U_{ij} \mathbf{e}_i \otimes \mathbf{e}_j, \quad T = T_{ij} \mathbf{e}_i \otimes \mathbf{e}_j.$$

Then

$$U_{ij} = C_{ijkl} T_{kl}. \quad (1.18)$$

Here i, j are free indices, while k, ℓ are dummy indices.



Proof.

We have

$$\begin{aligned} U_{ij} &= \mathbf{e}_i \cdot (U \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (\mathbf{C}(T) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (\mathbf{C}(T_{k\ell} \mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) T_{k\ell} \\ &= C_{ijkl} T_{k\ell}. \end{aligned}$$

The free indices i, j agree in every line, and k, ℓ are dummy indices. □



Example

Let $A \in \mathcal{V}^2$ be given and define $\mathbf{C} \in \mathcal{V}^4$ by

$$\mathbf{C}(T) = AT.$$

Then

$$\begin{aligned} C_{ijk\ell} &= \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot ((A\mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (A\mathbf{e}_k (\mathbf{e}_\ell \cdot \mathbf{e}_j)) \\ &= \delta_{\ell j} \mathbf{e}_i \cdot (A\mathbf{e}_k) \\ &= A_{ik} \delta_{\ell j}. \end{aligned}$$



Proposition

Let $\mathbf{C} \in \mathcal{V}^4$, and suppose $\mathcal{F} = \{\mathbf{e}_i\}$ is a frame. Then the components of $\mathbf{C}$ may equivalently be written

$$C_{ijkl} = (\mathbf{e}_i \otimes \mathbf{e}_j) : \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell).$$



Proof.

We first observe that, for any $A \in \mathcal{V}^2$,

$$A_{ij} = \mathbf{e}_i \cdot (A\mathbf{e}_j) = (\mathbf{e}_i \otimes \mathbf{e}_j) : A.$$

This follows from Result 1.13 (Lecture 04) because,

$$[\mathbf{e}_i \otimes \mathbf{e}_j]_{k\ell} = \delta_{ik}\delta_{j\ell}.$$



Proof (Cont.)

For fixed k, ℓ , define

$$A^{(k\ell)} := \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell) \in \mathcal{V}^2.$$

Applying the identity from the previous slide to $A^{(k\ell)}$ gives

$$\begin{aligned} C_{ijk\ell} &= \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (A^{(k\ell)} \mathbf{e}_j) \\ &= [A^{(k\ell)}]_{ij} \\ &= (\mathbf{e}_i \otimes \mathbf{e}_j) : A^{(k\ell)} \\ &= (\mathbf{e}_i \otimes \mathbf{e}_j) : \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell), \end{aligned}$$

which proves the result. □



Fourth-Order Dyadic Products

Fourth-Order Dyadic Products



Definition

Suppose that $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d} \in \mathcal{V}$. The object

$$\mathbf{a} \otimes \mathbf{b} \otimes \mathbf{c} \otimes \mathbf{d}$$

is a fourth-order tensor called the *dyadic product*. Its action is defined by

$$(\mathbf{a} \otimes \mathbf{b} \otimes \mathbf{c} \otimes \mathbf{d})T = (\mathbf{c} \cdot (T\mathbf{d}))\mathbf{a} \otimes \mathbf{b}, \quad T \in \mathcal{V}^2.$$



Proposition

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be a frame for $\mathcal{V}$. The set

$$\{\mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_\ell \mid 1 \leq i, j, k, \ell \leq 3\}$$

is a basis for $\mathcal{V}^4$. In particular, every $\mathbf{C} \in \mathcal{V}^4$ has the representation

$$\mathbf{C} = C_{ijkl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_\ell, \quad (1.19)$$

where

$$C_{ijkl} = \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell)\mathbf{e}_j).$$



Proof.

Let $\mathbf{C} \in \mathcal{V}^4$ be given. Suppose that $U, T \in \mathcal{V}^2$ satisfy

$$U = \mathbf{C}(T).$$

Write

$$U = U_{ij} \mathbf{e}_i \otimes \mathbf{e}_j, \quad T = T_{kl} \mathbf{e}_k \otimes \mathbf{e}_l.$$

Recall from equation (1.18) that

$$U_{ij} = C_{ijkl} T_{kl}.$$

Consequently,

$$U = C_{ijkl} T_{kl} \mathbf{e}_i \otimes \mathbf{e}_j.$$

Here i, j are free indices in the component equation, while k, l are dummy indices.



Proof (Cont.)

By the definition of the fourth-order dyadic product,

$$\begin{aligned} C_{ijkl}(\mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_\ell) T \\ &= C_{ijkl}(\mathbf{e}_k \cdot (T \mathbf{e}_\ell)) \mathbf{e}_i \otimes \mathbf{e}_j \\ &= C_{ijkl} T_{k\ell} \mathbf{e}_i \otimes \mathbf{e}_j \\ &= U \\ &= \mathbf{C}(T). \end{aligned}$$

Since this equality holds for every $T \in \mathcal{V}^2$,

$$C_{ijkl}(\mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_\ell) = \mathbf{C}.$$

Thus the displayed dyads span $\mathcal{V}^4$; uniqueness of the components gives linear independence. □



Symmetry Properties

Major Symmetry



Definition

A fourth-order tensor $\mathbf{C} \in \mathcal{V}^4$ is said to have *major symmetry* if and only if

$$A : \mathbf{C}(B) = \mathbf{C}(A) : B, \quad A, B \in \mathcal{V}^2.$$

Equivalently, $\mathbf{C}$ is self-adjoint with respect to the tensor inner product.

Minor Symmetries



Definition

A fourth-order tensor $\mathbf{C} \in \mathcal{V}^4$ is said to have *right minor symmetry* if and only if

$$A : \mathbf{C}(B) = A : \mathbf{C}(\text{sym}(B)), \quad A, B \in \mathcal{V}^2.$$

It is said to have *left minor symmetry* if and only if

$$A : \mathbf{C}(B) = \text{sym}(A) : \mathbf{C}(B), \quad A, B \in \mathcal{V}^2.$$



Result 1.14

If $\mathbf{C} \in \mathcal{V}^4$ has major symmetry, then

$$C_{ijkl} = C_{klij}, \quad i, j, k, \ell \in \{1, 2, 3\}.$$

If $\mathbf{C}$ has right and left minor symmetries, then, respectively,

$$C_{ijkl} = C_{ijek} \quad \text{and} \quad C_{ijkl} = C_{jilk}.$$



Proof.

Suppose that $\mathbf{C} \in \mathcal{V}^4$ has major symmetry. Then

$$\begin{aligned} C_{ijkl} &= \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) \\ &= (\mathbf{e}_i \otimes \mathbf{e}_j) : \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell) \\ &= \mathbf{C}(\mathbf{e}_i \otimes \mathbf{e}_j) : (\mathbf{e}_k \otimes \mathbf{e}_\ell) && \text{(major symmetry)} \\ &= (\mathbf{e}_k \otimes \mathbf{e}_\ell) : \mathbf{C}(\mathbf{e}_i \otimes \mathbf{e}_j) && \text{(symmetry of :)} \\ &= C_{k\ell ij}. \end{aligned}$$



Proof (Cont.)

The corresponding component calculations for the right and left minor symmetries give

$$C_{ijkl} = C_{ijlk} \quad \text{and} \quad C_{ijkl} = C_{jikl},$$

respectively.

These two computations are left as exercises.





Isotropic Tensor Functions



Isotropic Functions

Definition

A function $G : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ is called *isotropic* if and only if

$$QG(A)Q^T = G(QAQ^T)$$

for every $A \in \mathcal{V}^2$ and every rotation Q .

Definition

A function $g : \mathcal{V}^2 \rightarrow \mathbb{R}$ is called *isotropic* if and only if

$$g(A) = g(QAQ^T)$$

for every $A \in \mathcal{V}^2$ and every rotation Q .



Result 1.15

Let $G : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ be isotropic, with the property that $G(A)$ is symmetric whenever A is symmetric. Then there are functions

$$\alpha_0, \alpha_1, \alpha_2 : \mathbb{R}^3 \longrightarrow \mathbb{R}$$

such that

$$G(A) = \alpha_0(\mathcal{I}(A))I + \alpha_1(\mathcal{I}(A))A + \alpha_2(\mathcal{I}(A))A^2$$

for every symmetric $A \in \mathcal{V}^2$, where

$$\mathcal{I}(A) = (I_1(A), I_2(A), I_3(A)).$$



Result 1.15 (Cont.)

By Result 1.10, there are also functions

$$\beta_0, \beta_1, \beta_2 : \mathbb{R}^3 \longrightarrow \mathbb{R}$$

such that

$$G(A) = \beta_0(\mathcal{I}(A))I + \beta_1(\mathcal{I}(A))A + \beta_2(\mathcal{I}(A))A^{-1}$$

for every symmetric and invertible $A \in \mathcal{V}^2$.

Proof.

See M. E. Gurtin, *An Introduction to Continuum Mechanics*, Academic Press, 1981. □

Remark

Result 1.10 is the Cayley–Hamilton theorem.



Result 1.16

Let $\mathbf{C} \in \mathcal{V}^4$ be an isotropic fourth-order tensor with the properties

- ❶ $\mathbf{C}(A)$ is symmetric for every symmetric $A \in \mathcal{V}^2$;
- ❷ $\mathbf{C}(W) = O$ for every skew-symmetric $W \in \mathcal{V}^2$.

Then there are scalars μ and λ such that

$$\mathbf{C}(A) = \lambda \operatorname{tr}(A)I + 2\mu \operatorname{sym}(A), \quad A \in \mathcal{V}^2.$$



Proof.

Since $\mathbf{C} \in \mathcal{V}^4$ is isotropic and maps symmetric tensors to symmetric tensors, Result 1.15 gives functions

$$\alpha_0, \alpha_1, \alpha_2 : \mathbb{R}^3 \longrightarrow \mathbb{R}$$

such that

$$\mathbf{C}(H) = \alpha_0(\mathcal{I}(H))I + \alpha_1(\mathcal{I}(H))H + \alpha_2(\mathcal{I}(H))H^2$$

for every symmetric $H \in \mathcal{V}^2$, where

$$\mathcal{I}(H) = \left(\text{tr}(H), \frac{1}{2} [(\text{tr}(H))^2 - \text{tr}(H^2)], \det(H) \right).$$



Proof (Cont.)

A standard consequence of linearity and isotropy, obtained by evaluating $\mathbf{C}$ on diagonal symmetric tensors and using permutation invariance, is that there are constants $c_0, c_1, c_2 \in \mathbb{R}$ such that

$$\alpha_0(\mathcal{I}(H)) = c_0 \operatorname{tr}(H) + c_1, \quad \alpha_1(\mathcal{I}(H)) = c_2, \quad \alpha_2(\mathcal{I}(H)) = 0.$$

Therefore,

$$\mathbf{C}(H) = (c_0 \operatorname{tr}(H) + c_1)\mathbf{I} + c_2 H$$

for every symmetric $H \in \mathcal{V}^2$.



Proof (Cont.)

The zero tensor O is both symmetric and skew-symmetric. Hence condition (2) gives

$$\mathbf{C}(O) = O.$$

At the same time, the representation on the previous slide gives

$$\begin{aligned} O = \mathbf{C}(O) &= (c_0 \operatorname{tr}(O) + c_1)I + c_2 O \\ &= c_1 I. \end{aligned}$$

Thus $c_1 = 0$. Set

$$\lambda = c_0, \quad \mu = \frac{c_2}{2}.$$

Then

$$\mathbf{C}(H) = \lambda \operatorname{tr}(H)I + 2\mu H$$

for every symmetric $H \in \mathcal{V}^2$.



Proof (Cont.)

Let $A \in \mathcal{V}^2$ be arbitrary. Decompose it as

$$A = \text{sym}(A) + \text{skw}(A).$$

Since $\mathbf{C}$ is linear and condition (2) gives $\mathbf{C}(\text{skw}(A)) = O$,

$$\begin{aligned}\mathbf{C}(A) &= \mathbf{C}(\text{sym}(A)) + \mathbf{C}(\text{skw}(A)) \\ &= \lambda \text{tr}(\text{sym}(A))I + 2\mu \text{sym}(A) \\ &= \lambda \text{tr}(A)I + 2\mu \text{sym}(A).\end{aligned}$$

This holds for every $A \in \mathcal{V}^2$. □



Worked Exercises



Example (Exercise 1.7)

Given $\mathbf{a}, \mathbf{b} \in \mathcal{V}$ and $S \in \mathcal{V}^2$, prove that

①

$$S(\mathbf{a} \otimes \mathbf{b}) = (S\mathbf{a}) \otimes \mathbf{b};$$

②

$$(\mathbf{a} \otimes \mathbf{b})S = \mathbf{a} \otimes (S^T \mathbf{b});$$

③

$$(\mathbf{a} \otimes \mathbf{b})^T = \mathbf{b} \otimes \mathbf{a}.$$



Example (Exercise 1.7, Cont.)

For part (1), let $\mathbf{v} \in \mathcal{V}$ be arbitrary. Then

$$\begin{aligned}(S(\mathbf{a} \otimes \mathbf{b}))\mathbf{v} &= S((\mathbf{a} \otimes \mathbf{b})\mathbf{v}) \\ &= S((\mathbf{b} \cdot \mathbf{v})\mathbf{a}) \\ &= (\mathbf{b} \cdot \mathbf{v})S\mathbf{a} \\ &= ((S\mathbf{a}) \otimes \mathbf{b})\mathbf{v}.\end{aligned}$$

Since $\mathbf{v}$ is arbitrary,

$$S(\mathbf{a} \otimes \mathbf{b}) = (S\mathbf{a}) \otimes \mathbf{b}.$$



Example (Exercise 1.7, Cont.)

For part (2), let $\mathbf{v} \in \mathcal{V}$ be arbitrary. Then

$$\begin{aligned} ((\mathbf{a} \otimes \mathbf{b})S)\mathbf{v} &= (\mathbf{a} \otimes \mathbf{b})(S\mathbf{v}) \\ &= (\mathbf{b} \cdot (S\mathbf{v}))\mathbf{a} \\ &= (\mathbf{v} \cdot (S^T\mathbf{b}))\mathbf{a} \\ &= ((S^T\mathbf{b}) \cdot \mathbf{v})\mathbf{a} \\ &= (\mathbf{a} \otimes (S^T\mathbf{b}))\mathbf{v}. \end{aligned}$$

Since $\mathbf{v}$ is arbitrary,

$$(\mathbf{a} \otimes \mathbf{b})S = \mathbf{a} \otimes (S^T\mathbf{b}).$$



Example (Exercise 1.7, Cont.)

For part (3), let $\mathbf{w}, \mathbf{v} \in \mathcal{V}$ be arbitrary. Then

$$\begin{aligned}\mathbf{w} \cdot ((\mathbf{a} \otimes \mathbf{b})^T \mathbf{v}) &= \mathbf{v} \cdot ((\mathbf{a} \otimes \mathbf{b}) \mathbf{w}) \\ &= \mathbf{v} \cdot ((\mathbf{b} \cdot \mathbf{w}) \mathbf{a}) \\ &= (\mathbf{v} \cdot \mathbf{a})(\mathbf{b} \cdot \mathbf{w}).\end{aligned}$$

Choose $\mathbf{v} = \mathbf{e}_j$ and $\mathbf{w} = \mathbf{e}_i$. Then

$$[(\mathbf{a} \otimes \mathbf{b})^T]_{ij} = b_i a_j = [\mathbf{b} \otimes \mathbf{a}]_{ij}.$$

Thus

$$(\mathbf{a} \otimes \mathbf{b})^T = \mathbf{b} \otimes \mathbf{a}.$$



Example (Exercise 1.24)

Let $S \in \mathcal{V}^2$. Show that

$$I_2(S) = \frac{1}{2} [(\text{tr}(S))^2 - \text{tr}(S^2)]$$

has the same numerical value regardless of the frame in which it is calculated.

Suppose

$$\mathcal{F} = \{\mathbf{e}_i\} \quad \text{and} \quad \mathcal{F}' = \{\mathbf{e}'_i\}$$

are two frames, and $[S]$ and $[S]'$ are the matrix representations of S in these frames, respectively. By Result 1.5,

$$\text{tr}[S] = \text{tr}[S]'.$$



Example (Exercise 1.24, Cont.)

Let $[S^2]$ and $[S^2]'$ be the representations of S^2 in the two frames, and let A be the change-of-basis tensor from $\mathcal{F}$ to $\mathcal{F}'$. Then

$$[S^2]' = [A]^T [S^2] [A], \quad [S]' = [A]^T [S] [A].$$

Therefore,

$$\text{tr}[S^2] = \text{tr}[S^2]'.$$



Example (Exercise 1.24, Cont.)

Consequently,

$$\begin{aligned} I_2(S) &= \frac{1}{2} [(\text{tr}(S))^2 - \text{tr}(S^2)] \\ &= \frac{1}{2} [(\text{tr}[S])^2 - \text{tr}[S^2]] \\ &= \frac{1}{2} [(\text{tr}[S]')^2 - \text{tr}[S^2]'] . \end{aligned}$$

Thus $I_2(S)$ is frame-independent.



Example (Exercise 1.32)

Find the components of the fourth-order tensor $\mathbf{C} \in \mathcal{V}^4$ defined by

$$\mathbf{C}(S) = \frac{1}{2}(S - S^T).$$

By definition,

$$\begin{aligned} C_{ijkl} &= \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell)\mathbf{e}_j) \\ &= \mathbf{e}_i \cdot \left[\frac{1}{2} \{ (\mathbf{e}_k \otimes \mathbf{e}_\ell) - (\mathbf{e}_k \otimes \mathbf{e}_\ell)^T \} \mathbf{e}_j \right] \\ &= \mathbf{e}_i \cdot \left[\frac{1}{2} \{ (\mathbf{e}_k \otimes \mathbf{e}_\ell) - (\mathbf{e}_\ell \otimes \mathbf{e}_k) \} \mathbf{e}_j \right]. \end{aligned}$$



Example (Exercise 1.32, Cont.)

Continuing the component calculation,

$$\begin{aligned} C_{ijkl} &= \mathbf{e}_i \cdot \left[\frac{1}{2}(\mathbf{e}_\ell \cdot \mathbf{e}_j)\mathbf{e}_k - \frac{1}{2}(\mathbf{e}_k \cdot \mathbf{e}_j)\mathbf{e}_\ell \right] \\ &= \frac{1}{2} \mathbf{e}_i \cdot (\delta_{\ell j} \mathbf{e}_k - \delta_{kj} \mathbf{e}_\ell) \\ &= \frac{1}{2} \delta_{ik} \delta_{\ell j} - \frac{1}{2} \delta_{i\ell} \delta_{kj}. \end{aligned}$$

Thus the components are

$$C_{ijkl} = \frac{1}{2} (\delta_{ik} \delta_{\ell j} - \delta_{i\ell} \delta_{kj}).$$

Here i, j, k, ℓ are free indices throughout the component identity.